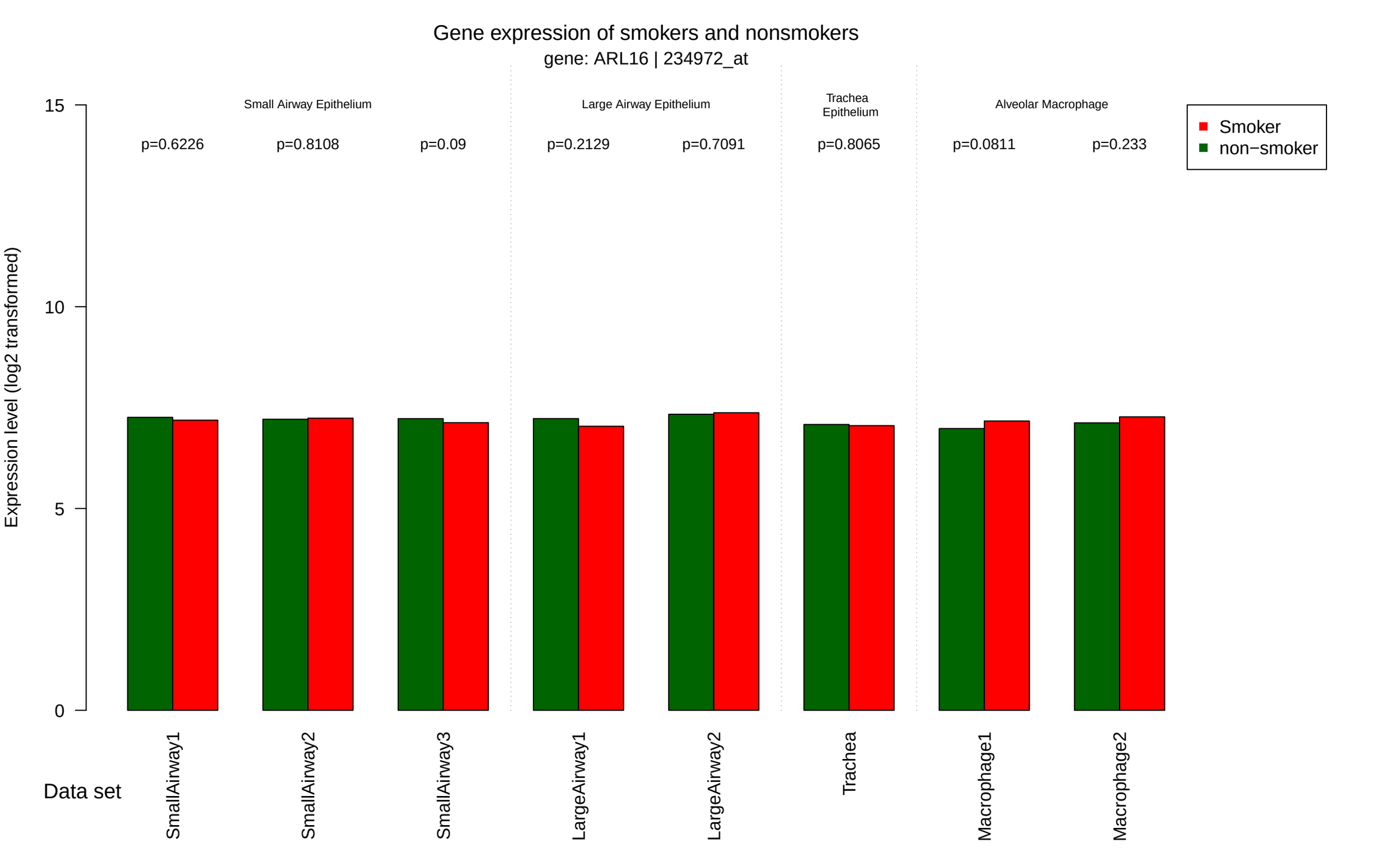
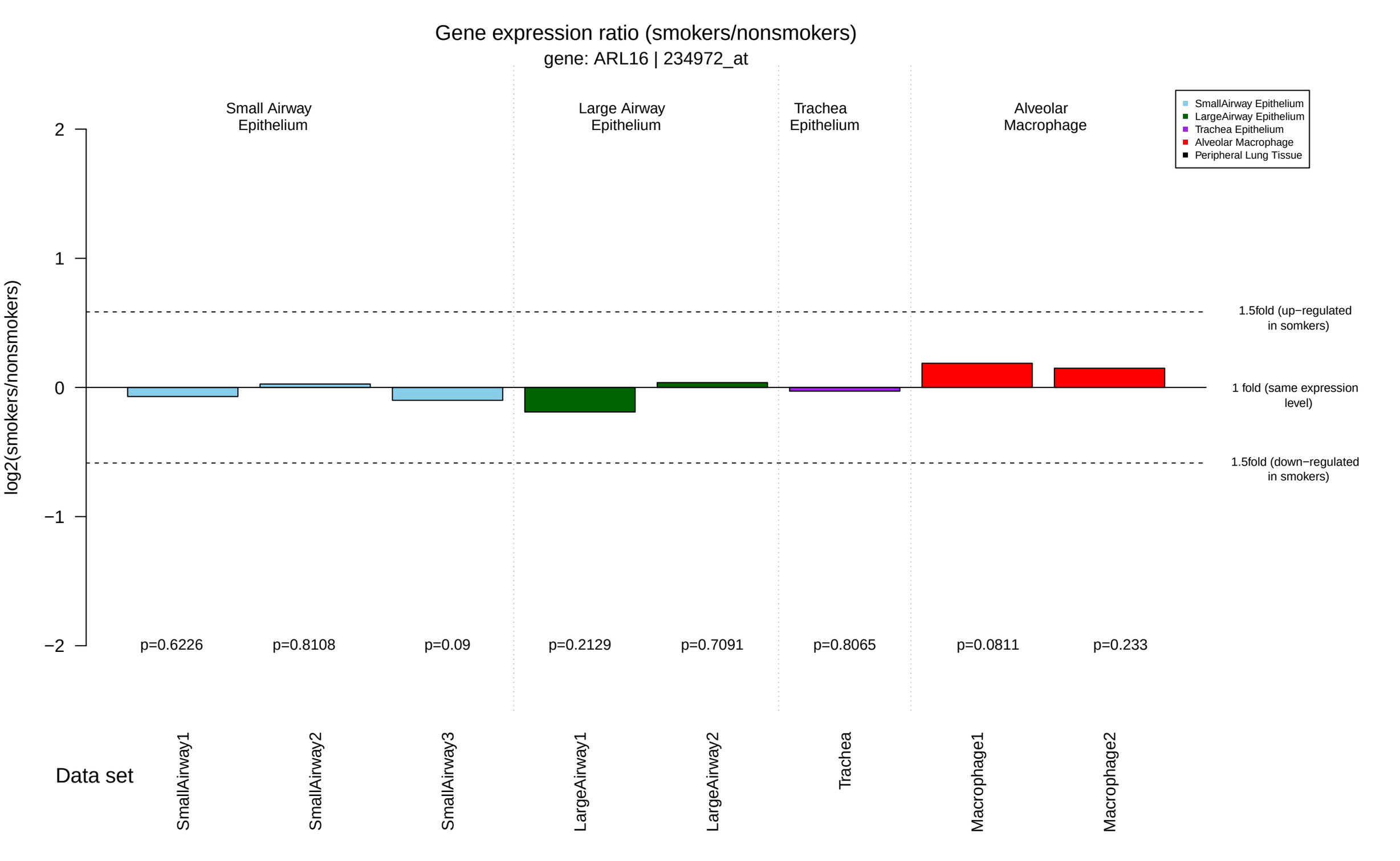
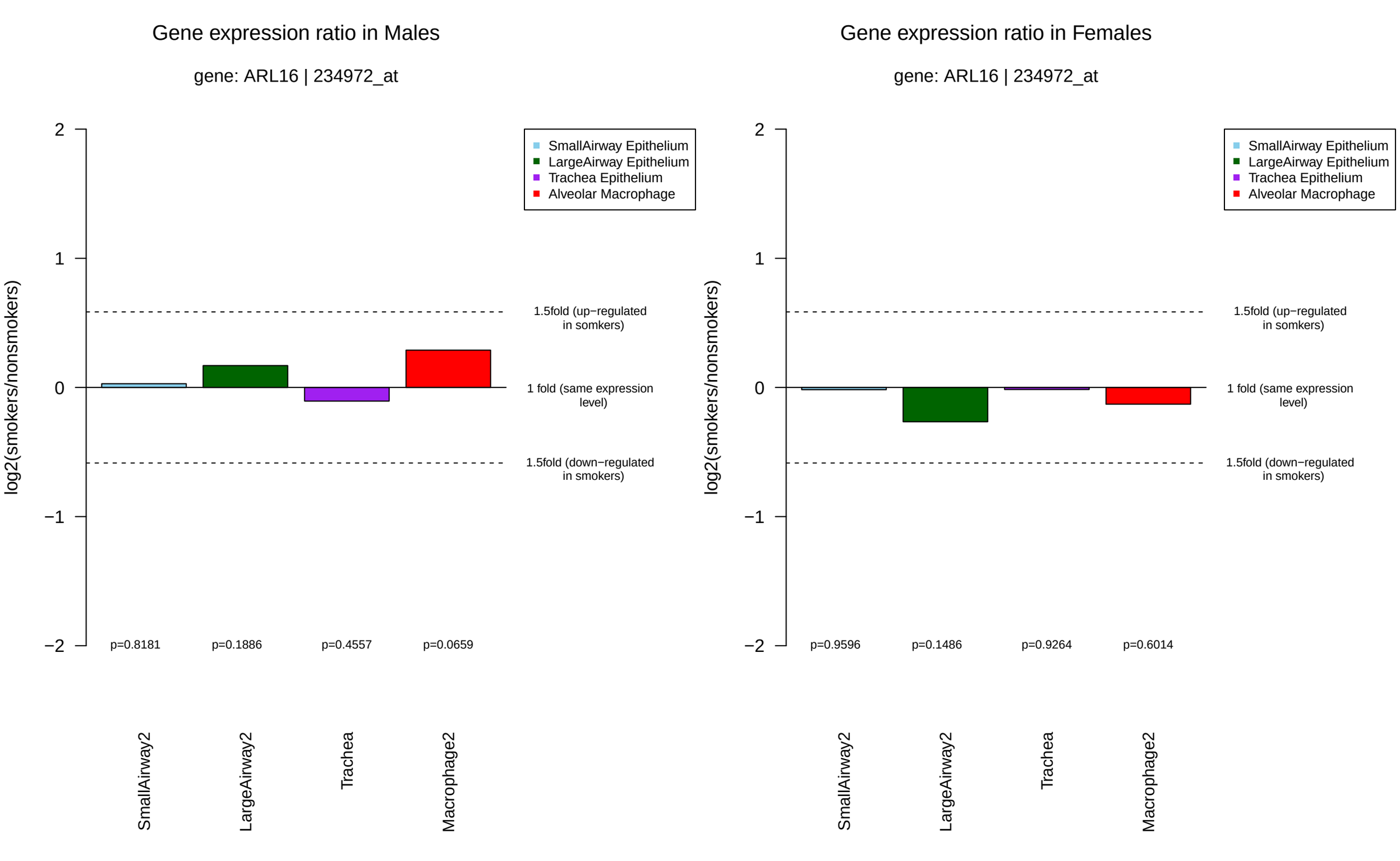
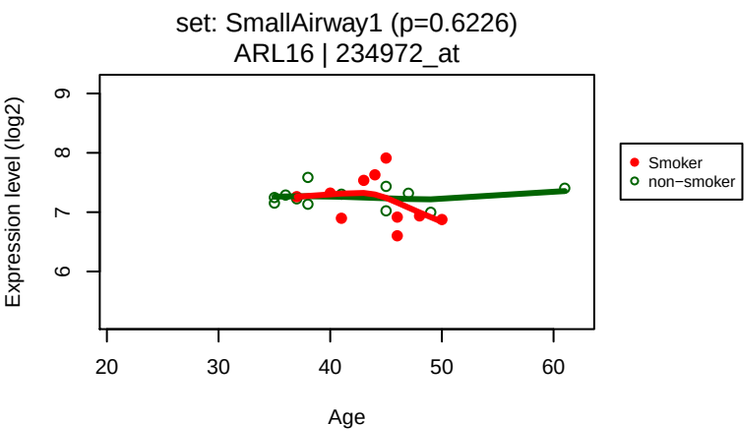
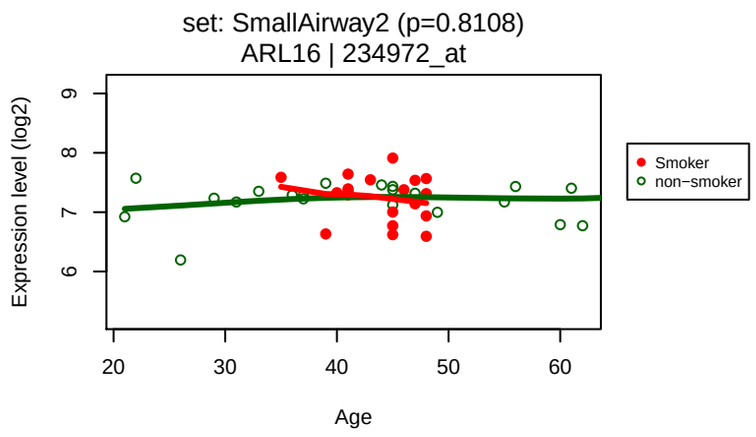
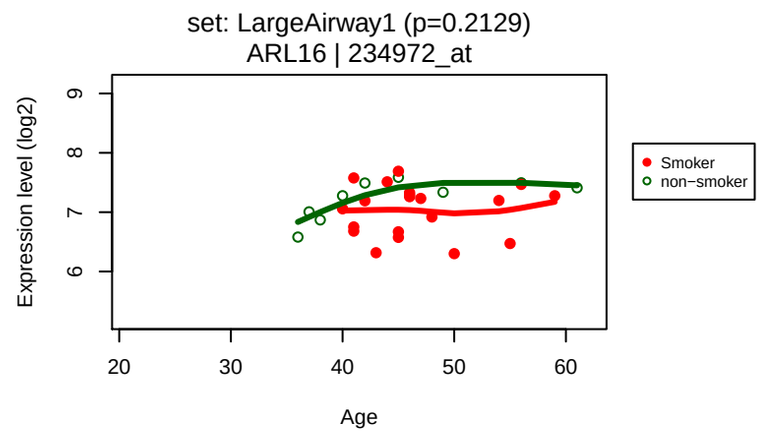

Gene expression of smokers and nonsmokers
gene: ARL16 | 234972_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6226
p=0.8108
p=0.09
p=0.2129
p=0.7091
p=0.8065
p=0.0811
p=0.233
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

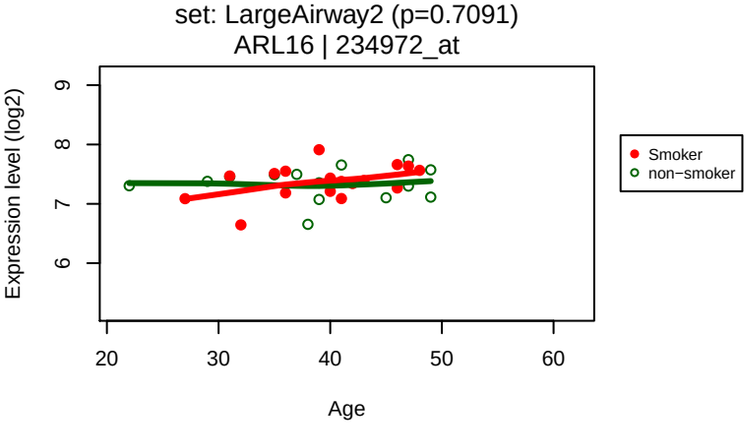

Gene expression ratio (smokers/nonsmokers)
gene: ARL16 | 234972_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6226
p=0.8108
p=0.09
p=0.2129
p=0.7091
p=0.8065
p=0.0811
p=0.233
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

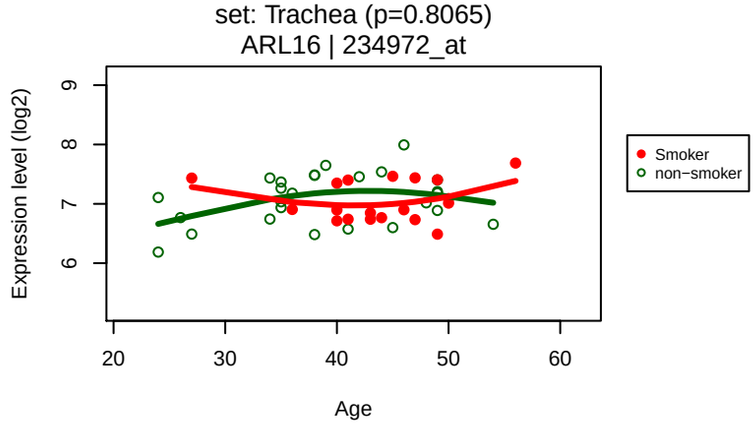

Gene expression ratio in Males
gene: ARL16 | 234972_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8181
p=0.1886
p=0.4557
p=0.0659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL16 | 234972_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9596
p=0.1486
p=0.9264
p=0.6014
SmallAirway2
LargeAirway2
Trachea
Macrophage2

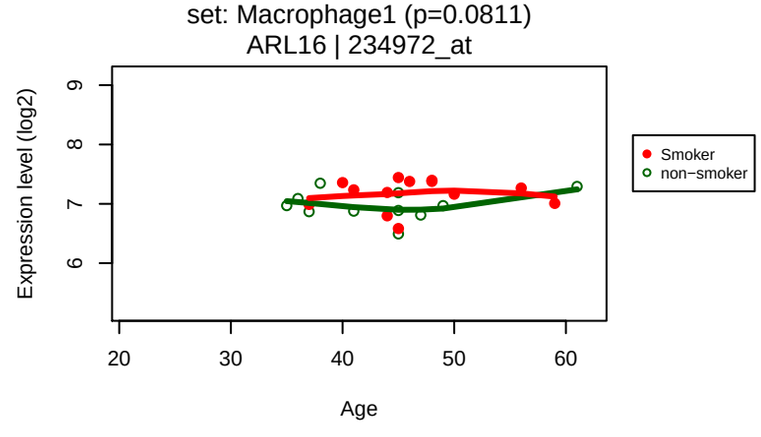

set: SmallAirway1 (p=0.6226)
ARL16 | 234972_at
Expression level (log2)
Age
Smoker
non−smoker

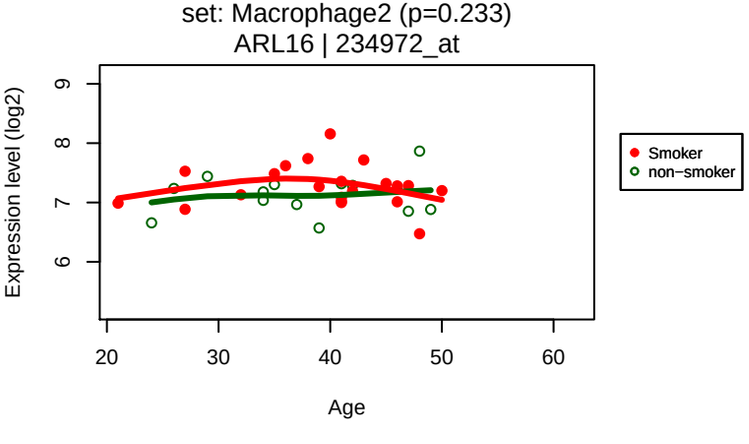

set: SmallAirway2 (p=0.8108)
ARL16 | 234972_at
Expression level (log2)
Age
Smoker
non−smoker

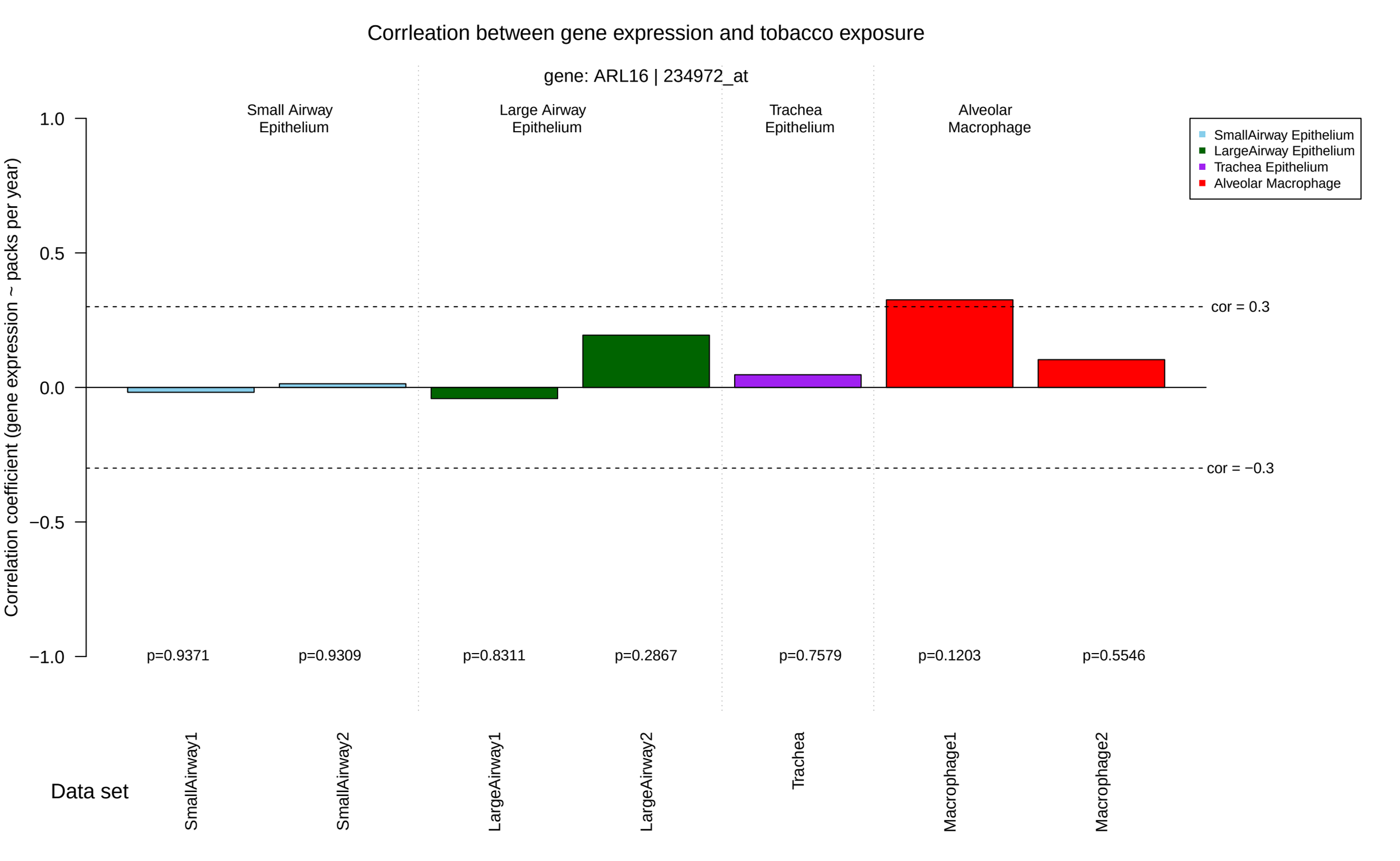

set: LargeAirway1 (p=0.2129)
ARL16 | 234972_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7091)
ARL16 | 234972_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8065)
ARL16 | 234972_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0811)
ARL16 | 234972_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.233)
ARL16 | 234972_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL16 | 234972_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9371
p=0.9309
p=0.8311
p=0.2867
p=0.7579
p=0.1203
p=0.5546
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2